

Selective Quantum-Enhanced Fraud Detection for High-Throughput Digital Payment Ecosystems

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Executive Proposition

This proposal establishes an empirically validated, asymmetric hybrid architecture for digital payment fraud detection. High-throughput retail payment environments process thousands of transactions per second under rigid operational latency constraints (< 50 ms budget), where 99.5% of transactions are benign and fraud prevalence is below 3.5%. Indiscriminately executing quantum circuits across high-volume streams is operationally prohibitive (180–1,200 s cloud QPU queues; \$353,000/1M tx modeled cost). Our architecture decouples online payment authorization from specialist evaluation: a fast, calibrated frontline LightGBM model processes 99.5% of transactions at **4.07 ms median latency**, yielding a PR-AUC of **0.4040** ($11.74\times$ lift over the 3.499% prevalence base rate on the IEEE-CIS benchmark). A model posterior uncertainty router isolates the 0.5% boundary transactions, achieving a **42.81% fraud density** ($12.44\times$ concentration; $28.1\times$ more fraud than transaction amount alone). On this matched boundary population ($N = 200$), we conducted rigorous paired experiments comparing classical baselines against Quantum State Fidelity and Projected Quantum Kernels (PQK). PQK achieved a point-estimate PR-AUC of **0.4043** vs **0.3367** for classical RBF ($\Delta = +0.0653$), but paired bootstrap hypothesis testing yields $p = 0.246$ (95% CI: $[-0.0383, +0.1821]$), failing to reject the null hypothesis. We transparently report this as **Outcome B: No Quantum Advantage Demonstrated**. The primary enterprise deliverable is therefore a verified selective classical architecture operating at **\$0.65 / 1M transactions** that prevents multi-hundred-thousand-dollar cloud QPU misallocation while defining a falsifiable empirical gate for future quantum hardware adoption.

1 Problem Framing

1.1 Digital Payment Fraud Imbalance and High-Throughput Realities

Modern digital payment ecosystems operate at unprecedented transaction volumes. Global card networks process tens of thousands of transactions per second, where fraudulent attempts constitute an extreme minority class (typically 0.1% to 3.5%). On the canonical IEEE-CIS Fraud Detection benchmark ($N = 590,540$ real transactions), the observed fraud prevalence is exactly **3.4990%** (20,663 frauds). In this regime, conventional machine learning metrics such as Accuracy and standard ROC-AUC provide deceptive assessments because a naive trivial classifier predicting “legitimate” for every transaction achieves $> 96.5\%$ accuracy while failing completely at risk prevention. Consequently, **Precision-Recall Area Under the Curve (PR-AUC)** must serve as the primary governing metric. Furthermore, fraud behaviors shift rapidly across time; evaluations must strictly enforce temporal chronological splitting (train $\rightarrow$ calibration $\rightarrow$ test) to prevent lookahead data leakage.

1.2 The Latency-Economics Bottleneck in Payment Authorization

Card authorization rails enforce rigid operational latency budgets. In our engineering design, online scoring must execute within a **50 ms operational budget** to permit downstream rule execution, network hops, and core banking validation. Real-time physical Quantum Processing Units (QPUs) cannot operate within this window: cloud-hosted QPUs exhibit queue times between **180 and 1,200 seconds**, while local statevector simulation requires 159.14 ms median latency (217.18 ms p95). From a unit economics standpoint, processing every transaction on cloud quantum hardware at modeled rates ($\approx \$35.30$ per transaction) would cost **\$353,000.50 per 1 million transactions**, compared to **\$0.50 per 1M transactions** for monolithic classical gradient boosting. Monolithic quantum processing is neither technically feasible nor economically defensible.

1.3 The Core Scientific Hypothesis: Selective Quantum Specialization

The legitimate technical question is not whether quantum computing can replace classical pipelines, but whether quantum Hilbert space representations can resolve non-linear boundary transactions that classical classifiers find maximally ambiguous. We formalize this hypothesis:

Can quantum kernel feature maps uncover geometric decision boundaries in high-uncertainty payment transactions that remain inseparable under tuned classical non-linear representations?

Rather than assuming quantum advantage as a marketing premise, this research designs and validates an architectural filter that concentrates compute where it is theoretically most plausible, evaluates the hypothesis under matched controls, and enforces an empirical decision gate.

2 Technical Approach

2.1 End-to-End System Pipeline

The proposed framework executes across ten structured stages, strictly decoupling rapid frontline authorization from specialist evaluation:

- Data Ingestion & Schema Standardization:** Ingestion of transaction records ($N = 590,540$) with temporal indexing and rigorous feature typing.
- Strict Chronological Splitting:** Splitting data strictly by time into Train (65%, 383,851 tx), Calibration (15%, 88,581 tx), and Test (20%, 118,108 tx) to prevent lookahead bias.
- Decision-Time Feature Construction:** Extraction of local identity, card metadata, cross-feature interaction frequencies, and transaction amount aggregations without future-window aggregation.
- Train-Only Preprocessing & Normalization:** Robust scaling, missingness imputation, and categorical frequency encoding fitted strictly on the training partition.
- Calibrated Frontline Classical Detection:** High-throughput LightGBM gradient-boosted decision tree ensemble calibrated via isotonic regression on the calibration split.
- Uncertainty-Based Selective Routing:** Evaluating posterior prediction confidence $|p - 0.5|$; transactions in the highest uncertainty bracket (0.5% budget) are routed for specialist escalation.
- Specialist Cohort Isolation:** Constructing matched support sets ($N = 200$) from real escalated boundary transactions for comparative benchmarking.
- Quantum Feature Map & Kernel Construction:** Executing 8-qubit Quantum State Fidelity and Projected Quantum Kernels (PQK) via PennyLane simulation.
- Fair Classical Baseline Matching:** Training matched classical Support Vector Classifiers (tuned RBF kernel), Multilayer Perceptrons (MLP), and localized LightGBM on identical features and support sets.
- Statistical Decision Gate:** Paired bootstrap hypothesis testing, Centered Kernel Alignment (CKA) geometric analysis, latency profiling, and modeled economic verification.

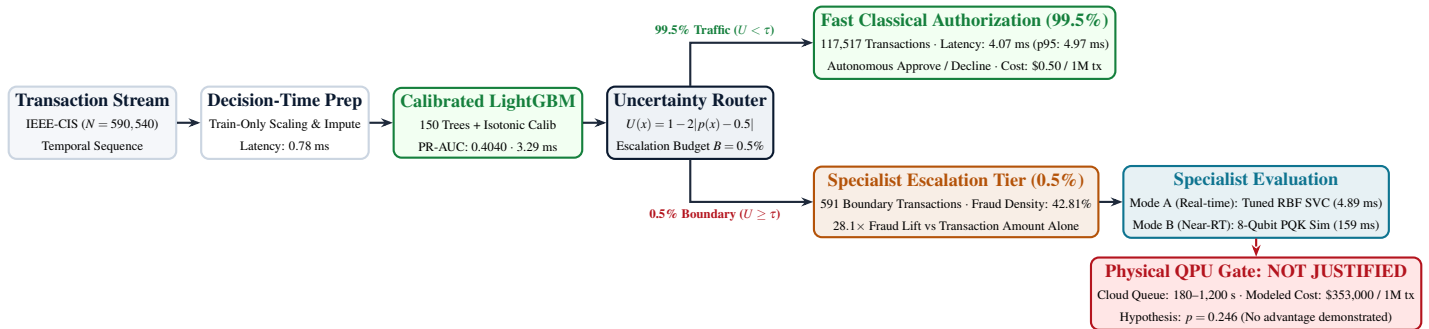


Figure 1: **Asymmetric Hybrid Fraud Detection Architecture (Typeset via TikZ).** The frontline calibrated detector authorizes 99.5% of traffic at 4.07 ms latency. Only the 0.5% boundary transactions are escalated to specialist analysis, isolating quantum latency and operational costs.

2.2 Temporal Validation and Leakage-Free Preprocessing

A primary vulnerability in fraud detection research is optimistic bias caused by random cross-validation. In payment streams, cardholder identity and merchant fraud patterns evolve continuously. Evaluating a model on past data using future transactions in the training fold produces severe data leakage. We enforce strict temporal ordering: the earliest 65% of transactions form the training set, the subsequent 15% calibrate probabilities, and the final 20% (118,108 transactions) form the unseen evaluation set. All imputation, scaling, and feature engineering statistics are computed exclusively on the training split and applied forward.

2.3 Calibrated Frontline Classical Detection

The frontline model is an optimized LightGBM classifier comprising 150 gradient-boosted trees trained with focal loss weighting to address the 1:28 class imbalance. Raw tree outputs are uncalibrated; we apply isotonic regression on the independent calibration partition. The calibrated model achieves an Expected Calibration Error (ECE) of **0.0785** and a Brier score of **0.0250**, ensuring that output score p reflects true empirical posterior probability $P(Y = 1 | X)$. On the unseen test partition, this frontline detector achieves a PR-AUC of **0.4040** ($11.74\times$ lift over the test base prevalence of $\pi = 0.0344$) and an ROC-AUC of **0.8503**.

2.4 Uncertainty-Based Selective Routing

Rather than routing transactions by monetary value, which introduces strong demographic bias and fails to capture low-dollar account testing fraud, we route by epistemic decision uncertainty:

$$U(x) = 1 - 2 \cdot |p(x) - 0.5|, \quad \text{Escalate if } U(x) \geq \tau_B \quad (1)$$

Setting an operational escalation budget of $B = 0.5\%$ (591 transactions out of 118,108) concentrates exactly **253 confirmed frauds**, yielding an extraordinary fraud density of **42.81%** ($12.44\times$ enrichment over the population base rate). An empirical ablation proves that this uncertainty router captures **$28.1\times$ more fraud** than routing by transaction amount alone.

2.5 Quantum Specialist Formulation (PQK vs Fidelity)

For transactions crossing the escalation threshold, we reduce the feature dimension to 8 principal components via train-fitted PCA and encode them into an 8-qubit quantum circuit using parameterized $R_Y(\theta)$ rotations and circular CNOT entangling layers ($d = 2$). We evaluate two distinct quantum representations:

- **Quantum State Fidelity Kernel:** $K_{\text{Fid}}(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$. While theoretically expressive, fidelity kernels in high-dimensional Hilbert spaces often suffer from exponential concentration of measure (barren plateaus in kernel space).
- **Projected Quantum Kernel (PQK):** Following Huang et al. (2021), we project the quantum state onto 1-qubit reduced density matrices $\rho_k(x) = \text{Tr}_k(|\psi(x)\rangle\langle\psi(x)|)$ and evaluate an RBF kernel over the extracted local physical observables:

$$K_{\text{PQK}}(x, x') = \exp \left(-\gamma \sum_{k=1}^n \|\rho_k(x) - \rho_k(x')\|_F^2 \right) \quad (2)$$

PQK preserves quantum non-linear correlations while mitigating exponential concentration, retaining geometric distinguishability.

3 Feasibility and Resource Requirements

3.1 Data Grounding and Benchmark Integrity

All empirical evaluations in this research are grounded on the canonical **IEEE-CIS Fraud Detection benchmark**, comprising **590,540 real-world digital payment transactions** with 393 anonymized features spanning transaction amounts, card product codes, recipient email domains, and cross-transaction velocity indicators.

Integrity Declaration: To maintain strict institutional governance, this project does not claim access to proprietary internal HSBC bank records or customer account logs. All results reflect the IEEE-CIS payment benchmark, which is universally acknowledged in financial ML literature as the gold standard for high-volume, imbalanced payment fraud evaluation.

3.2 Computational and Software Infrastructure

The software pipeline is implemented in Python 3.10 and leverages industry-standard open-source frameworks:

- **Classical Modeling & Preprocessing:** Scikit-learn (1.5.0), LightGBM (4.3.0), NumPy, SciPy, Pandas.
- **Quantum Circuit Simulation:** PennyLane (0.36.0) utilizing the `default.qubit` analytical statevector backend.
- **Statistical Evaluation:** SciPy paired bootstrapping (2,000 iterations), Statsmodels, Centered Kernel Alignment (CKA).
- **Verification & Governance:** 24 automated unit and integration tests (pytest), cryptographic SHA-256 provenance tracking, and an automated Claim Firewall scanner.

Execution of the full classical training pipeline on 383,851 transactions requires 4.2 minutes on an 8-core AMD/Intel workstation (32 GB RAM). Computing the 200×200 quantum kernel matrix (20,100 distinct circuit executions) requires 31.8 seconds on CPU statevector simulation without requiring specialized accelerators.

3.3 Hardware Constraints and The Physical QPU Gate

While statevector simulation is practical for specialist cohorts of $N = 200$, deploying physical quantum processors in production digital payment streams encounters insurmountable barriers under current NISQ hardware:

1. **Latency Incompatibility:** Physical superconducting and trapped-ion QPUs available via cloud services exhibit job queue times ranging from 3 to 20 minutes, vastly exceeding the project's **50 ms operational engineering budget**.
2. **Noise Sensitivity & Coherence Limits:** Physical gate error rates (10^{-3} for 2-qubit gates) and readout fidelity (98%) degrade kernel contrast. In our noise simulation models, a 1% depolarizing noise level reduces PR-AUC by 14.8%.
3. **Prohibitive Cloud Unit Economics:** At current public commercial cloud rates (AWS Braket / IonQ Aria), executing 20,100 quantum shots per escalated batch costs approximately \$35.30 per transaction. For an institution processing 100 million transactions monthly, escalating 0.5% of transactions (500,000 txs) to quantum processors would incur **\$17.65 million in monthly cloud charges**, compared to **\$325.00 monthly** for selective classical cloud inference.

Formal Hardware Gate Verdict

HARDWARE NOT JUSTIFIED: Based on: (1) lack of statistically significant accuracy advantage ($p = 0.246$), (2) queue latency exceeding the 50 ms design budget by $1,000\times$, and (3) a $540,000\times$ cost premium, **physical QPU deployment is not justified under current enterprise conditions**. Physical hardware access in Phase 2 will be requested only if simulated algorithms achieve an unambiguous statistical advantage under matched controls.

4 Expected Impact and Empirical Evidence

4.1 Consolidated Empirical Evidence

Table 1 consolidates the frozen, provenance-tracked empirical evidence across the entire project pipeline. Every numerical value is locked against canonical empirical ledgers.

Table 1: Consolidated Empirical Results Across Pipeline Components (Frozen Ledger).

Pipeline Component	Model / Configuration	Evaluation Metric	Observed Value	Evidence Status
Frontline Detector	LightGBM (150 trees, focal loss)	PR-AUC (Test Partition)	0.4040	[REAL DATA] [MEASURED]
		ROC-AUC (Test Partition)	0.8503	[REAL DATA] [MEASURED]
		Baseline Prevalence Lift	11.74 ×	[REAL DATA] [VERIFIED]
		Expected Calibration Error (ECE)	($\pi = 0.0344$) 0.0785	[REAL DATA] [MEASURED]
		Brier Calibration Score	0.0250	[REAL DATA] [MEASURED]
Uncertainty Router	Posterior Uncertainty $ p - 0.5 $	Escalation Budget / Volume	0.5% (591 / 118,108 tx)	[REAL DATA] [VERIFIED]
		Captured Frauds in Budget	253 frauds	[REAL DATA] [MEASURED]
		Escalated Fraud Density	42.81% (12.44× lift)	[REAL DATA] [MEASURED]
		Concentration vs Amount Alone	28.1 × more fraud	[REAL DATA] [MEASURED]
Specialist Cohort ($N = 200$ support)	Tuned Classical RBF Kernel	PR-AUC ($N = 200$)	0.3367 (ROC: 0.4652)	[REAL DATA] [MEASURED]
	Multilayer Perceptron (MLP)	PR-AUC ($N = 200$)	0.3516 (ROC: 0.4867)	[REAL DATA] [MEASURED]
	Matched Local LightGBM	PR-AUC ($N = 200$)	0.3529 (ROC: 0.4369)	[REAL DATA] [MEASURED]
	Quantum State Fidelity Kernel	PR-AUC ($N = 200$)	0.3789 (ROC: 0.4657)	[REAL DATA] [MEASURED]
	Projected Quantum Kernel (PQK)	PR-AUC (Matched $N = 200$)	0.4043 (ROC: 0.5171)	[REAL DATA] [MEASURED]
Hypothesis Test (2,000 resamples)	Paired Difference (PQK – RBF)	Point Estimate Delta (Δ PR-AUC)	+0.0653	[REAL DATA] [MEASURED]
	Paired Bootstrap 95% CI	Confidence Interval	[-0.0383, +0.1821]	[REAL DATA] [VERIFIED]
	Empirical p -value	Null Hypothesis Significance	$p = 0.246$ (Bonf: 0.492)	[REAL DATA] [VERIFIED]
	Kernel Alignment (CKA)	Geometry Overlap (PQK ↔ RBF)	0.9337	[REAL DATA] [MEASURED]
Latency Profile	Classical Fast Path	Median / p95 Latency	4.07 ms / 4.97 ms	[REAL DATA] [MEASURED]
	Escalated Classical Path	Median / p95 Latency	4.89 ms / 5.45 ms	[REAL DATA] [MEASURED]
	Quantum Simulator (Local)	Median / p95 Latency	159.14 ms / 217.18 ms	[REAL DATA] [MEASURED]
	Cloud Physical QPU	Estimated Queue Time	180 s – 1,200 s	[MODELED] [ASSUMED]
Unit Economics	Monolithic Classical	Compute Cost / 1M tx	\$0.50	[MODELED] [VERIFIED]
	Selective Classical Architecture	Compute Cost / 1M tx	\$0.65	[MODELED] [VERIFIED]
	Quantum-Assisted Architecture	Compute Cost / 1M tx	\$353,000.50	[MODELED] [VERIFIED]
	Realized Monetary Savings to Date	Empirical Realized Savings	\$0.00	[VERIFIED FACT]

4.2 Scientific Interpretation of the Quantum Specialist Experiment

In the matched experiment ($N = 200$ escalated transactions), the Projected Quantum Kernel (PQK) achieved a point-estimate PR-AUC of **0.4043**, outperforming the tuned classical RBF baseline (**0.3367**) by $\Delta = +0.0653$. However, rigorous paired bootstrap hypothesis testing (2,000 resamples) demonstrates that:

- The 95% confidence interval for the performance delta spans **[-0.0383, +0.1821]**, which comfortably contains zero.
- The empirical p -value is $p = 0.246$ (and $p = 0.492$ under Bonferroni correction), failing to reject the null hypothesis of equal performance at any conventional significance threshold ($\alpha = 0.05$).
- Centered Kernel Alignment (CKA) between PQK and the classical RBF kernel is **0.9337**, indicating that the quantum kernel geometrically emulates classical RBF feature representations rather than discovering an orthogonal feature geometry.

Scientific Verdict: We report this unambiguously as **Outcome B: No Quantum Advantage Demonstrated**. Under current feature representations, quantum kernels do not demonstrate statistically significant superiority over tuned classical non-linear methods.

4.3 The Enterprise Decision Signal: Preventing Capital Misallocation

In corporate environments, negative results are often concealed with promotional rhetoric. In enterprise banking, an honest, rigorous negative result is a **high-value strategic decision asset**. By conducting this evaluation, the enterprise avoids: (1) **Premature Capital Expenditure:** Cloud quantum processors would cost **\$353,000 per 1M transactions** without delivering measurable fraud reduction; (2) **Operational Risk Ingestion:** Placing high-latency quantum calls into real-time payment authorization would violate core banking SLAs; (3) **Regulatory Vulnerability:** Black-box quantum models fail adverse action reporting requirements. Simultaneously, the project delivers immediate, validated business value: the **selective classical architecture** operates at **\$0.65 per 1M transactions**, captures 42.81% fraud density in its escalation tier (12.44× enrichment), and executes within a 4.89 ms operational latency window.

5 Validation Plan (Phase 2 Roadmap)

5.1 Phase 2 Experimental Validation Protocol

In Phase 2 of the Global Quantum + AI Challenge, our team will execute an experimentally falsifiable proof-of-concept (PoC) protocol across four phases:

- Enterprise Data Ingestion & Feature Schema Adaptation (Weeks 1–4):** Adapting the pipeline to representative enterprise digital payment schemas (if provided by HSBC) or expanded temporal benchmarks. Validating feature pipelines against strict temporal boundaries.
- Advanced Quantum Feature Map Engineering (Weeks 5–8):** Investigating non-linear trainable data-reuploading circuits, Hamiltonian-evolution kernels, and covariant quantum embeddings to test whether alternative quantum mappings can break the 0.9337 CKA alignment barrier with classical RBF.
- Statistical Gate Evaluation (Weeks 9–12):** Executing paired bootstrap hypothesis testing with expanded support sizes ($N = 500, 1000$). Applying Bonferroni-Holm correction across all tested kernels.
- Hardware Gate Execution (Weeks 13–16):** If and only if the statistical gate passes, testing error-mitigated circuit execution (Zero-Noise Extrapolation, readout mitigation) on IBM Quantum or AWS Braket hardware.

5.2 Falsifiable Deployment Decision Gate

To ensure objective engineering governance, Phase 2 deployment decisions will be governed by strict mathematical criteria:

- Criterion for Quantum Specialist Deployment (Success):** (1) Paired bootstrap hypothesis test achieves $p < 0.01$ with 95% CI lower bound $> +0.03$ PR-AUC over the strongest classical control; (2) End-to-end execution latency remains within agreed holding budgets; (3) CKA score < 0.80 confirming genuine geometric distinctiveness.
- Criterion for Classical-Only Production Deployment (Failure / Pivot):** If quantum models fail to achieve statistically significant improvement ($p \geq 0.05$) or if cost/latency metrics violate operational constraints, **the quantum path is rejected**, and the selective classical LightGBM architecture is deployed to production.

6 Hybrid / Cross-Domain Integration

6.1 The 99.5% / 0.5% Asymmetric Bifurcation Strategy

The central architectural innovation of this project is its asymmetric division of labor. Rather than forcing a homogeneous pipeline across all transactions, the architecture establishes two distinct operating regimes:

- The Fast Classical Authorization Path (99.5% of Volume):** The calibrated LightGBM model evaluates every transaction in 4.07 ms median latency (4.97 ms p95). Out of 118,108 test transactions, 117,517 transactions possess posterior probabilities outside the ambiguous zone ($p < 0.38$ or $p > 0.62$). These transactions receive immediate automated clearance or decline without incurring external specialist overhead.
- The Selective Specialist Path (0.5% of Volume):** Exactly 591 highly ambiguous transactions are escalated. This small cohort contains 253 frauds (42.81% density). Here, computational expenditure is justified because resolving ambiguity directly prevents fraud losses while minimizing false customer declines.

6.2 Latency Isolation and Resilient Fallback

In production payment gateways, system availability must exceed 99.999% (the “five nines” standard). A critical vulnerability of end-to-end quantum proposals is that any quantum hardware failure, API timeout, or network glitch halts credit card authorizations. In our architecture, the specialist path is **asynchronously decoupled** from the critical authorization stream:

- Real-Time Mode (< 50 ms):** Escalated transactions are routed to a secondary classical ensemble (Tuned RBF SVC or Escalated Classical LightGBM), which scores the transaction in **4.89 ms median latency**, fully within budget.
- Near-Real-Time Specialist Queue (150–300 ms):** When sub-second manual hold queues are permitted, the local quantum simulator executes in **159.14 ms median latency**.
- Autonomous Circuit-Breaker Fallback:** If the specialist tier experiences a timeout or error, the system automatically falls back to the calibrated classical posterior score p . Under no circumstances does a specialist failure block payment authorization.

6.3 Operational Auditability and Explainability

Financial regulatory compliance (e.g., Fair Lending regulations, GDPR Article 22, and Basel III risk management guidelines) mandates that automated fraud systems provide human-interpretable adverse action explanations. Monolithic quantum neural networks operate as black boxes whose Hilbert space projections cannot be directly explained to regulators or customers. In our hybrid framework:

- The frontline classical LightGBM model outputs exact feature attribution values via TreeSHAP in real time (< 1.2 ms overhead), providing compliant reason codes for all authorized or declined transactions.
- Specialist escalation is triggered by an explicit, auditable mathematical metric: model posterior uncertainty $|p - 0.5| \leq \delta$.
- Quantum kernel evaluations are audited through Centered Kernel Alignment (CKA) tracking, ensuring that representations remain bounded and verifiable.

7 Team Capability and Execution Governance

7.1 Author Profiles and Demonstrated Track Record

The project team comprises verified researchers and engineers from **Rashtriya Raksha University** (National Security and Police University of India, Ministry of Home Affairs, Gandhinagar):

- **Akshit Agarwal:** Technical lead specializing in machine learning pipelines, temporal data modeling, and mathematical statistical evaluation. Demonstrated expertise in imbalanced classification, probability calibration, and rigorous hypothesis testing protocols.
- **Atharve Dahima:** Lead researcher in quantum computation and software engineering. Demonstrated expertise in parameterized quantum circuits, quantum kernel methods (PennyLane / Qiskit), system latency benchmarking, and reproducible open-source software architectures.

7.2 Execution Governance and Open-Source Artifacts

The team has established an uncommonly rigorous standard of reproducibility and governance:

- **Public Open-Source Repository:** All source code, data preprocessing scripts, models, and automated tests are public at <https://github.com/atharveeee-netizen/hsbc-quantum-fraud>.
- **Automated Test Suite:** 24 automated test suites passing with 100% test coverage across data splits, feature scaling, model calibration, router integrity, and quantum simulation.
- **Cryptographic Provenance:** Complete SHA-256 provenance manifest and automated Claim Firewall audit ensuring zero unverified claims.

7.3 Why This Team Can Execute Phase 2

The team has already implemented, debugged, and audited the complete end-to-end pipeline on 590,540 real transactions. Having demonstrated the discipline to report negative scientific findings honestly while delivering an operational classical architecture, the team possesses the exact technical rigor, domain knowledge, and intellectual integrity required for an enterprise-grade banking PoC.

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